

Texas Geometry Credit by Exam

Practice Exam A · 22 Questions

McGraw Hill Glencoe Geometry (Texas) 2015 · TEKS Aligned

Name: _____	Date: _____
Score: _____ / 22	Time Allowed: _____ min

DIRECTIONS: Select the best answer for each question. Each question is worth 1 point. Show all work in the space below each question. Calculators may be used unless otherwise noted. This exam is aligned to the Texas Essential Knowledge and Skills (TEKS) for Geometry.

[Logic – Conditional Statements]

1. What is the converse of the statement:

"If two angles are vertical angles, then they are congruent."

- A. If two angles are congruent, then they are vertical angles.
- B. If two angles are not vertical angles, then they are not congruent.
- C. If two angles are not congruent, then they are not vertical angles.
- D. If two angles are vertical angles, then they are supplementary.

Work space: _____

[Logic – Conditional Statements]

2. What is the contrapositive of:

"If it is raining, then the ground is wet."

- A. If the ground is wet, then it is raining.
- B. If it is not raining, then the ground is not wet.
- C. If the ground is not wet, then it is not raining.
- D. If the ground is wet, then it is not raining.

Work space: _____

[Angle Relationships – Supplementary]

3. Two angles are supplementary. One angle measures $(3x + 15)^\circ$ and the other measures $(2x - 5)^\circ$. What is the value of x ?

- A. 30
- B. 34
- C. 38
- D. 40

Work space: _____

[Parallel Lines and Transversals]

4. Two parallel lines are cut by a transversal. One co-interior (consecutive interior / same-side interior) angle measures 65° . What is the measure of the other co-interior angle?

- A. 25°
- B. 65°
- C. 115°
- D. 125°

Work space: _____

[Triangle Congruence]

5. Which congruence shortcut applies when two sides and the included angle of one triangle are congruent to those of another?

- A. SSS
- B. SAS
- C. ASA
- D. AAS

Work space: _____

[Exterior Angle Theorem]

6. An exterior angle of a triangle measures 110° . One non-adjacent interior angle (remote interior angle) measures 42° . What is the measure of the other remote interior angle?

- A. 48°
- B. 68°
- C. 70°
- D. 75°

Work space: _____

[Pythagorean Theorem]

7. What is the length of the hypotenuse of a right triangle with legs measuring 5 and 12?

- A. 11
- B. 13
- C. 15
- D. 17

Work space: _____

[Special Right Triangles (30-60-90)]

8. In a 30° - 60° - 90° triangle, the shorter leg measures 7. What is the length of the hypotenuse?

- A. $7\sqrt{2}$
- B. $7\sqrt{3}$
- C. 14
- D. $14\sqrt{3}$

Work space: _____

[Coordinate Geometry – Distance Formula]

9. What is the distance between the points (1, 2) and (4, 6)?

- A. 4
- B. 5
- C. 6
- D. 7

Work space: _____

[Coordinate Geometry – Midpoint Formula]

10. What is the midpoint of the segment with endpoints $(-2, 5)$ and $(8, -3)$?

- A. (3, 1)
- B. (5, 4)
- C. (3, 4)
- D. (6, 2)

Work space: _____

[Coordinate Geometry – Slope]

11. What is the slope of a line perpendicular to $y = \frac{3}{4}x + 2$?

- A. $\frac{3}{4}$
- B. $\frac{4}{3}$
- C. $-\frac{4}{3}$
- D. $-\frac{3}{4}$

Work space: _____

[Triangle Midsegment Theorem]

12. A midsegment of a triangle connects the midpoints of two sides and has length 18 cm. What is the length of the side of the triangle parallel to this midsegment?

- A. 9 cm
- B. 18 cm
- C. 27 cm
- D. 36 cm

Work space: _____

[Triangle Inequality Theorem]

13. Which set of side lengths could represent the sides of a triangle?

- A. 3, 4, 9
- B. 5, 5, 10
- C. 6, 8, 11
- D. 2, 7, 10

Work space: _____

[Polygon Interior Angle Sum]

14. What is the sum of the interior angles of a hexagon?

- A. 540°
- B. 720°
- C. 900°
- D. 1080°

Work space: _____

[Parallel Lines – Converse Theorems]

15. Two lines are cut by a transversal. The alternate interior angles are congruent. Which conclusion is valid?

- A. The lines are perpendicular.
- B. The lines are parallel.
- C. The lines are skew.
- D. The lines intersect at a 45° angle.

Work space: _____

[Coordinate Geometry – Linear Equations]

16. A line passes through the point (0, 3) with slope 2. What is the equation of the line?

- A. $y = 2x - 3$
- B. $y = 3x + 2$
- C. $y = 2x + 3$
- D. $y = -2x + 3$

Work space: _____

[Isosceles Triangle Theorem]

17. In an isosceles triangle, the vertex angle measures 40° . What is the measure of each base angle?

- A. 60°
- B. 70°
- C. 80°
- D. 140°

Work space: _____

[Angle Bisector]

18. Ray BD bisects angle ABC. $m\angle ABD = (3x + 5)^\circ$ and $m\angle DBC = (5x - 7)^\circ$. What is $m\angle ABC$?

- A. 6°
- B. 23°
- C. 46°
- D. 51°

Work space: _____

[Segment Addition Postulate]

19. Point B is between A and C. $AB = (3x - 2)$, $BC = (x + 8)$, and $AC = 30$. What is x ?

- A. 5
- B. 6
- C. 7
- D. 8

Work space: _____

[Angle Addition Postulate]

20. Point D is in the interior of $\angle ABC$. $m\angle ABD = (2x + 10)^\circ$ and $m\angle DBC = (3x - 5)^\circ$. If $m\angle ABC = 70^\circ$, what is x ?

- A. 11
- B. 13
- C. 15
- D. 17

Work space: _____

[Converse of the Pythagorean Theorem]

21. A triangle has sides of length 8, 15, and 17. Which type of triangle is it?

- A. Acute triangle
- B. Obtuse triangle
- C. Right triangle
- D. Equilateral triangle

Work space: _____

[Geometric Mean]

22. What is the geometric mean of 4 and 16?

- A. 6
- B. 8
- C. 10
- D. 12

Work space: _____

Answer Key – Practice Exam A

#	Answer	#	Answer
1	A	12	D
2	C	13	C
3	B	14	B
4	C	15	B
5	B	16	C
6	B	17	B
7	B	18	C
8	C	19	B
9	B	20	B
10	A	21	C
11	C	22	B

TEKS Standards Covered

Q #	Topic	TEKS
1	Conditional Statements – Converse	G.4(B)
2	Conditional Statements – Contrapositive	G.4(B)
3	Supplementary Angles	G.6(A)
4	Parallel Lines – Co-interior Angles	G.5(A)
5	Triangle Congruence – SAS	G.6(B)
6	Exterior Angle Theorem	G.6(D)
7	Pythagorean Theorem	G.9(B)
8	30-60-90 Special Right Triangle	G.9(B)
9	Distance Formula	G.2(B)
10	Midpoint Formula	G.2(B)
11	Perpendicular Slopes	G.2(B)
12	Triangle Midsegment Theorem	G.6(D)
13	Triangle Inequality Theorem	G.6(D)
14	Polygon Interior Angle Sum	G.6(A)
15	Converse – Alternate Interior Angles	G.5(A)
16	Linear Equations – Slope-Intercept	G.2(B)
17	Isosceles Triangle Theorem	G.6(D)
18	Angle Bisector	G.6(A)

19	Segment Addition Postulate	G.2(A)
20	Angle Addition Postulate	G.6(A)
21	Converse of Pythagorean Theorem	G.9(B)
22	Geometric Mean	G.9(B)

Detailed Explanations – Exam A

1. (A) [Logic – Conditional Statements]

The **converse** of "If P, then Q" is "If Q, then P" — swap hypothesis and conclusion. Converse: "If two angles are congruent, then they are vertical angles." (Note: the converse is not always true — congruent angles are not necessarily vertical angles.)

2. (C) [Logic – Conditional Statements]

The **contrapositive** of "If P, then Q" is "If NOT Q, then NOT P." Negate and swap both parts: "If the ground is NOT wet, then it is NOT raining." The contrapositive is always logically equivalent to the original.

3. (B) [Angle Relationships – Supplementary]

Supplementary angles sum to 180° :

$$(3x + 15) + (2x - 5) = 180 \rightarrow 5x + 10 = 180 \rightarrow 5x = 170 \rightarrow x = 34.$$

$$\text{Check: } (3 \cdot 34 + 15) + (2 \cdot 34 - 5) = 117 + 63 = 180. \checkmark$$

4. (C) [Parallel Lines and Transversals]

Co-interior (same-side interior) angles are **supplementary** when lines are parallel. $180^\circ - 65^\circ = 115^\circ$.

5. (B) [Triangle Congruence]

SAS (Side-Angle-Side): two sides and the angle *between* them (the included angle) are congruent. The key word is 'included' — the angle must be between the two congruent sides.

6. (B) [Exterior Angle Theorem]

By the **Exterior Angle Theorem**, an exterior angle equals the sum of the two remote interior angles: $110 = 42 + x \rightarrow x = 68^\circ$.

7. (B) [Pythagorean Theorem]

$$a^2 + b^2 = c^2 \rightarrow 5^2 + 12^2 = c^2 \rightarrow 25 + 144 = 169 \rightarrow c = \sqrt{169} = 13.$$

(5, 12, 13) is a classic Pythagorean triple.

8. (C) [Special Right Triangles (30-60-90)]

In a 30-60-90 triangle the sides are in ratio $1 : \sqrt{3} : 2$.

Shorter leg = 1 unit $\rightarrow$ hypotenuse = 2 units. If shorter leg = 7, then hypotenuse = $2 \times 7 = 14$.

9. (B) [Coordinate Geometry – Distance Formula]

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

10. (A) [Coordinate Geometry – Midpoint Formula]

$$\text{Midpoint} = ((x_1 + x_2)/2, (y_1 + y_2)/2) = ((-2 + 8)/2, (5 + (-3))/2) = (6/2, 2/2) = (3, 1).$$

11. (C) [Coordinate Geometry – Slope]

Perpendicular lines have slopes that are **negative reciprocals**. The slope of the given line is $3/4$. Its negative reciprocal is $-4/3$.

12. (D) [Triangle Midsegment Theorem]

By the **Triangle Midsegment Theorem**, the midsegment is parallel to the third side and half its length. So the parallel side = $2 \times 18 = 36$ cm.

13. (C) [Triangle Inequality Theorem]

The **Triangle Inequality Theorem** states that the sum of any two sides must be greater than the third side.

$$6 + 8 = 14 > 11 \checkmark, 6 + 11 = 17 > 8 \checkmark, 8 + 11 = 19 > 6 \checkmark$$

The others fail: $3 + 4 = 7$ (not > 9); $5 + 5 = 10$ (not > 10); $2 + 7 = 9$ (not > 10).

14. (B) [Polygon Interior Angle Sum]

Interior angle sum = $(n - 2) \times 180^\circ$.

For a hexagon ($n = 6$): $(6 - 2) \times 180^\circ = 4 \times 180^\circ = 720^\circ$.

15. (B) [Parallel Lines – Converse Theorems]

By the **Converse of the Alternate Interior Angles Theorem**: if alternate interior angles are congruent, the lines are parallel.

16. (C) [Coordinate Geometry – Linear Equations]

Slope-intercept form: $y = mx + b$.

$m = 2$, $b = 3$ (y-intercept at $(0, 3)$). Equation: $y = 2x + 3$.

17. (B) [Isosceles Triangle Theorem]

The two base angles are equal. Let each base angle = b .

$$40 + 2b = 180 \rightarrow 2b = 140 \rightarrow b = 70^\circ.$$

18. (C) [Angle Bisector]

Since BD bisects $\angle ABC$, the two halves are equal:

$$3x + 5 = 5x - 7 \rightarrow 12 = 2x \rightarrow x = 6.$$

$$m\angle ABD = 3(6) + 5 = 23^\circ. m\angle ABC = 2 \times 23^\circ = 46^\circ.$$

19. (B) [Segment Addition Postulate]

$$AB + BC = AC \rightarrow (3x - 2) + (x + 8) = 30 \rightarrow 4x + 6 = 30 \rightarrow 4x = 24 \rightarrow x = 6.$$

$$\text{Check: } AB = 16, BC = 14, 16 + 14 = 30 \checkmark$$

20. (B) [Angle Addition Postulate]

$$m\angle ABD + m\angle DBC = m\angle ABC:$$

$$(2x + 10) + (3x - 5) = 70 \rightarrow 5x + 5 = 70 \rightarrow 5x = 65 \rightarrow x = 13.$$

21. (C) [Converse of the Pythagorean Theorem]

$$\text{Test: } 8^2 + 15^2 \text{ vs } 17^2$$

$64 + 225 = 289 = 17^2$. Since $a^2 + b^2 = c^2$, this is a **right triangle** (Converse of Pythagorean Theorem).

22. (B) [Geometric Mean]

$$\text{Geometric mean of } a \text{ and } b = \sqrt{a \cdot b} = \sqrt{4 \times 16} = \sqrt{64} = 8.$$

$$\text{Equivalently: } 4/x = x/16 \rightarrow x^2 = 64 \rightarrow x = 8.$$